

Compounds **1** and **2** are best described as:

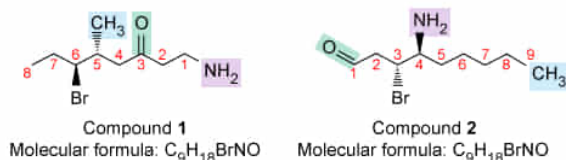
- ☐ A. diastereomers. [3%]
- ☐ B. enantiomers. [1%]
- ✓ ☒ C. structural isomers. [81%]
- ☐ D. conformational isomers. [13%]

Correct

81% answered correctly

### Explanation:

#### Structural isomers



- Same molecular formula
- $CH_3$ ,  $C=O$ , and  $NH_2$  groups are in different locations

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Isomers are molecules with the same molecular formula and can be **classified** based on the connectivity (eg, **constitutional isomers**) and spatial orientation (eg, stereoisomers) of the atoms in their molecular structures. Constitutional isomers (also called **structural isomers**) differ in how the atoms are connected at one or more points within the structure.

Compounds **1** and **2** have the same molecular formula, but some of the **functional groups** are in different locations along the primary carbon chain in each structure:

- The methyl group on carbon 5 in Compound **1** is located at the end of the carbon chain in Compound **2**.
- The carbonyl (ie, the  $C=O$  bond) incorporated in the **ketone** in Compound **1** appears at the end of the carbon chain in Compound **2** as part of an **aldehyde**.
- The **amine** on carbon 1 in Compound **1** is located on carbon 4 in Compound **2**.

Therefore, Compounds **1** and **2** are structural isomers.

(Choices **A**, **B**, and **D**) **Diastereomers**, **enantiomers**, and **conformational isomers** are

stereoisomers with the *same* atom connectivity but different spatial orientation. Because Compounds **1** and **2** do not have the same atom connectivity, they cannot be any of these types of stereoisomers.

**Educational objective:**

Isomers are molecules with the same molecular formula and can be classified based on connectivity and spatial orientation. Structural isomers, also called constitutional isomers, differ in how the atoms are connected at one or more points within the structure.

**Time spent:**0 sec

**Subject:**

Organic Chemistry

**QID:**403236

**System:**

Introduction to Organic Chemistry